

CME 222

MASS TRANSFER

A hand holding a small, round, gold-colored perfume bottle. From the bottle, numerous thin, pink, curved lines radiate outwards to the left, illustrating the concept of diffusive mass transfer. The background is a light, warm gradient.

Diffusive Mass Transfer

Dr. Tan Peng Chee
A4 #407
pengchee.tan@xmu.edu.my

Table of Content

- Molecular Diffusivity
- Estimation of Diffusion Coefficient (Gas Diffusivity)

Molecular Diffusivity

Diffusion Coefficient

Dimension

$$D_{AB} = \frac{-J_{A,z}}{dc_A/dz} = \left(\frac{N}{L^2 t} \right) \left(\frac{1}{N/L^3 \cdot 1/L} \right) = \frac{L^2}{t} \quad \text{Unit: cm}^2/\text{s or m}^2/\text{s}$$

- Depends on the pressure, temperature, composition of system, medium of diffusion
- Diffusion coefficients/mass diffusivity

For **gases**: 10^{-6} to 10^{-5} m²/s

For **liquids**: 10^{-10} to 10^{-9} m²/s,

For **solids**: 10^{-14} to 10^{-10} m²/s

Diffusion Coefficient

Table J.1 Binary mass diffusivities in gases[†]

System	T (K)	$D_{AB}P$ (cm ² atm/s)	$D_{AB}P$ (m ² Pa/s)
Air			
Ammonia	273	0.198	2.006
Aniline	298	0.0726	0.735
Benzene	298	0.0962	0.974
Bromine	293	0.091	0.923
Carbon dioxide	273	0.136	1.378
Carbon disulfide	273	0.0883	0.894
Chlorine	273	0.124	1.256
Diphenyl	491	0.160	1.621
Ethyl acetate	273	0.0709	0.718
Ethanol	298	0.132	1.337
Ethyl ether	293	0.0896	0.908
Iodine	298	0.0834	0.845
Methanol	298	0.162	1.641
Mercury	614	0.473	4.791
Naphthalene	298	0.0611	0.619
Nitrobenzene	298	0.0868	0.879
<i>n</i> -Octane	298	0.0602	0.610
Oxygen	273	0.175	1.773
Propyl acetate	315	0.092	0.932
Sulfur dioxide	273	0.122	1.236
Toluene	298	0.0844	0.855
Water	298	0.260	2.634
Ammonia			
Ethylene	293	0.177	1.793
Argon			
Neon	293	0.329	3.333
Carbon dioxide			
Benzene	318	0.0715	0.724
Carbon disulfide	318	0.0715	0.724

Diffusion Coefficient

System	T (K)	$D_{AB}P$ (cm ² atm/s)	$D_{AB}P$ (m ² Pa/s)
Ethanol	273	0.0693	0.702
Ethyl ether	273	0.0541	0.548
Hydrogen	273	0.550	5.572
Methane	273	0.153	1.550
Methanol	298.6	0.105	1.064
Nitrogen	298	0.165	1.672
Nitrous oxide	298	0.117	1.185
Propane	298	0.0863	0.874
Water	298	0.164	1.661
Carbon monoxide			
Ethylene	273	0.151	1.530
Hydrogen	273	0.651	6.595
Nitrogen	288	0.192	1.945
Oxygen	273	0.185	1.874
Helium			
Argon	273	0.641	6.493
Benzene	298	0.384	3.890
Ethanol	298	0.494	5.004
Hydrogen	293	1.64	16.613
Neon	293	1.23	12.460
Water	298	0.908	9.198
Hydrogen			
Ammonia	293	0.849	8.600
Argon	293	0.770	7.800
Benzene	273	0.317	3.211
Ethane	273	0.439	4.447
Methane	273	0.625	6.331
Oxygen	273	0.697	7.061
Water	293	0.850	8.611
Nitrogen			
Ammonia	293	0.241	2.441
Ethylene	298	0.163	1.651
Hydrogen	288	0.743	7.527
Iodine	273	0.070	0.709
Oxygen	273	0.181	1.834
Oxygen			
Ammonia	293	0.253	2.563
Benzene	296	0.0939	0.951
Ethylene	293	0.182	1.844

Diffusion Coefficient

Table J.2 Binary mass diffusivities in liquids[†]

Solute <i>A</i>	Solvent <i>B</i>	Temperature (K)	Solute concentration (g mol/L or kg mol/m ³)	Diffusivity (cm ² /s × 10 ⁵ or m ² /s × 10 ⁹)
Chlorine	Water	289	0.12	1.26
Hydrogen chloride	Water	273	9	2.7
			2	1.8
		283	9	3.3
			2.5	2.5
Ammonia	Water	289	0.5	2.44
		278	3.5	1.24
		288	1.0	1.77
Carbon dioxide	Water	283	0	1.46
		293	0	1.77
Sodium chloride	Water	291	0.05	1.26
			0.2	1.21
			1.0	1.24
			3.0	1.36
			5.4	1.54
Methanol	Water	288	0	1.28
Acetic acid	Water	285.5	1.0	0.82
			0.01	0.91
			1.0	0.96
Ethanol	Water	291	1.0	0.96
		283	3.75	0.50
		289	0.05	0.83
			2.0	0.90
<i>n</i> -Butanol	Water	288	0	0.77
Carbon dioxide	Ethanol	290	0	3.2
Chloroform	Ethanol	293	2.0	1.25

Diffusion Coefficient

Table J.3 Binary diffusivities in solids[†]

Solute	Solid	Temperature (K)	Diffusivity (cm ² /s or m ² /s × 10 ⁴)	Diffusivity (ft ² /h)
Helium	Pyrex	293	4.49×10^{-11}	1.74×10^{-10}
		773	2.00×10^{-8}	7.76×10^{-8}
Hydrogen	Nickel	358	1.16×10^{-8}	4.5×10^{-8}
		438	1.05×10^{-7}	4.07×10^{-7}
Bismuth	Lead	293	1.10×10^{-16}	4.27×10^{-16}
Mercury	Lead	293	2.50×10^{-15}	9.7×10^{-15}
Antimony	Silver	293	3.51×10^{-21}	1.36×10^{-20}
Aluminum	Copper	293	1.30×10^{-30}	5.04×10^{-30}
Cadmium	Copper	293	2.71×10^{-15}	1.05×10^{-14}

Diffusion Coefficient

Equality of Diffusion Coefficient

- Only true for a binary system of **constant molar density** (concentration)
- Occur at **constant temperature and pressure** for gas system
- One mole of A move in z direction, one mole of B move in opposite direction

$$c_A + c_B = c = \frac{P}{RT}$$

$$J_A = -J_B$$

$$-D_{AB} \frac{dc_A}{dz} = -(-D_{BA} \frac{dc_B}{dz})$$

$$D_{AB} = D_{BA}$$

Estimation of Diffusion Coefficient (Gas Diffusivity)

Gas Mass Diffusivity

- Based upon the kinetic theory of gases
- Assumption:
 - ✓ Ideal gas
 - ✓ Low density gaseous mixture
 - ✓ Molecules are regarded as rigid spheres that exert no intermolecular forces
 - ✓ Elastic collision between the rigid molecules

Gas Mass Diffusivity

Self-Diffusion Coefficient

- Diffusivity for an ideal gas mixture of species A diffusing through its isotope A_{*}

$$D_{AA^*} = \frac{1}{3} \lambda u$$

Mean Free Path (λ)

$$\lambda = \frac{\kappa T}{\sqrt{2} \pi \sigma_A^2 P}$$

Mean Speed (u)

$$u = \sqrt{\frac{8 \kappa N T}{\pi M_A}}$$

Gas Mass Diffusivity

Substituting the expression into first equation

$$D_{AA*} = \frac{2T^{\frac{3}{2}}}{3\pi^{\frac{3}{2}}\sigma_A^2 P} \left(\frac{\kappa^3 N}{M_A} \right)^{\frac{1}{2}}$$

M_A : Molecular weight of diffusing species A (kg/mol)

N : Avogadro's number (6.022×10^{23} molecules/mol)

P : Pressure (Pa)

T : Absolute temperature (K)

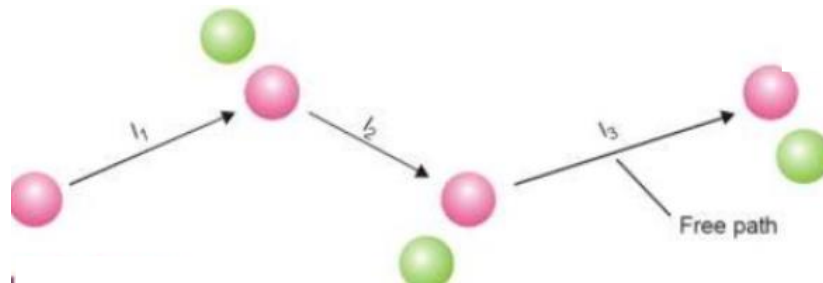
κ : Boltzmann constant (1.38×10^{-23} J/K)

σ : Lennard-Jones diameter of the spherical molecules (m)

Gas Mass Diffusivity

Mean Free Path

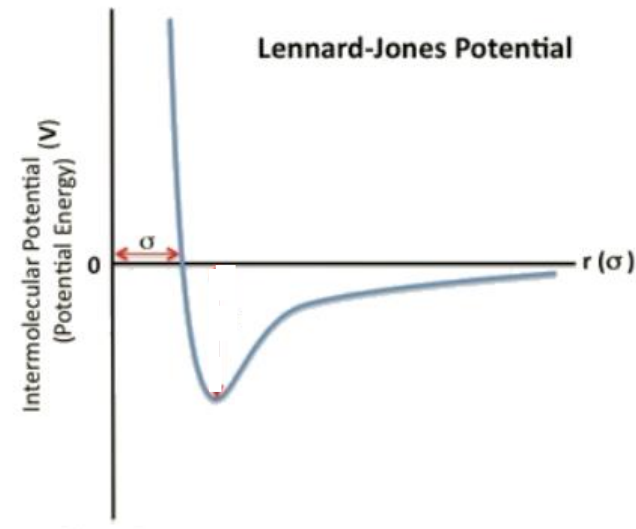
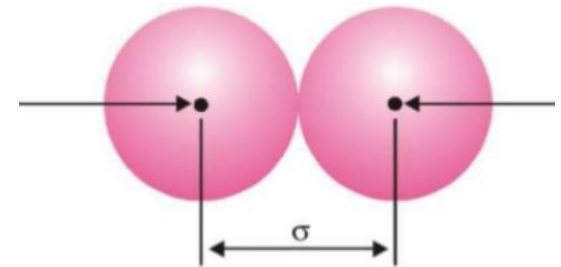
- A molecule travels in a straight line before collision
- The distance travelled by a molecule before collision is termed free path
- Mean free path: **average distance** travelled by a molecule **between** two successive **collision**
- Mean free path increases as the number of molecules per unit volume decreases (less collision)



Gas Mass Diffusivity

Lennard-Jones Diameter/Collision Diameter

- When two gas molecules approach one another, they cannot come closer beyond a certain distance
- Is the **closest distance** between the centers of the two molecules taking part in a collision
- Is the distance at which the **intermolecular potential** between the two particles is **zero**
- Related to the mean free path: smaller the collision or molecular diameter, the larger the mean free path



Gas Mass Diffusivity

Diffusion Coefficient for Binary Mixture

- A and B with unequal diameters

$$D_{AB} = \frac{2}{3} \left(\frac{\kappa}{\pi} \right)^{\frac{3}{2}} N^{\frac{1}{2}} T^{\frac{3}{2}} \frac{\left(\frac{1}{2M_A} + \frac{1}{2M_B} \right)^{\frac{1}{2}}}{P \left(\frac{\sigma_A + \sigma_B}{2} \right)^2}$$

$$D_{AB} = D_{BA}$$

Gas Mass Diffusivity

- Gas phase diffusion is dependent on pressure and temperature

$$D_{AB} \propto \frac{1}{P} \qquad D_{AB} \propto T^{3/2}$$

- Can use to predict mass diffusivity at different pressure and temperature

Gas Mass Diffusivity

Hirschfelder Equation

- A more rigorous equation
- Account for **force of attraction and repulsion** between molecules
- Estimate diffusion coefficient for gas pairs of **nonpolar and nonreacting** molecules

$$D_{AB} = \frac{0.001858 T^{\frac{3}{2}} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{\frac{1}{2}}}{P \sigma_{AB}^2 \Omega_D}$$

D_{AB} : mass diffusivity of A through B (cm²/s)

T: absolute temperature (K)

M_A, M_B : molecular weight (g/mol)

P: absolute pressure (atm)

σ_{AB} : collision diameter (Å)

Ω_D : collision integral (dimensionless)

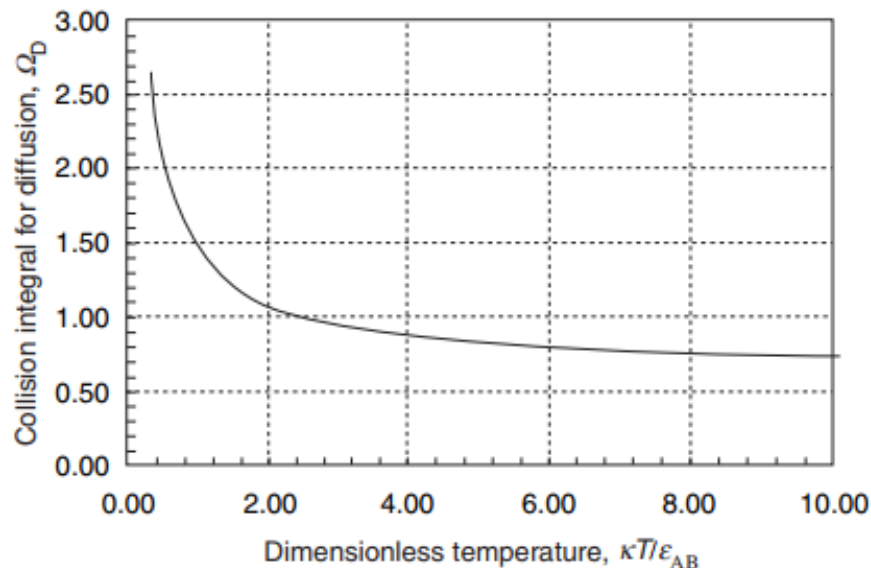
$$\sigma_{AB} = \frac{\sigma_A + \sigma_B}{2}$$

$$\varepsilon_{AB} = \sqrt{\varepsilon_A \varepsilon_B}$$

Gas Mass Diffusivity

Collision Integral

- Is a function of dimensionless temperature $\frac{\kappa T}{\varepsilon_{AB}}$
- ε_{AB} is the energy of molecular interaction for binary system A and B (J)



Binary gas phase Lennard-Jones collision integral

Gas Mass Diffusivity

Hirschfelder Equation

- Can be used to extrapolate experimental data
- Collision integral (Ω_D) decreases with increasing temperature, causing increase of D_{AB}
- So, D_{AB} influence by P, T and Ω_D

$$D_{ABT_2,P_2} = D_{ABT_1,P_1} \left(\frac{P_1}{P_2} \right) \left(\frac{T_2}{T_1} \right)^{\frac{3}{2}} \frac{\Omega_{D,T_1}}{\Omega_{D,T_2}}$$

Example 1

- a. Evaluate the diffusion coefficient of carbon dioxide in air at 20°C and atmospheric pressure.
- b. Compare this value with the experimental value reported in appendix table J.1.

Gas Mass Diffusivity

Fuller Correlation

- Used when reliable Lennard-Jones parameter (σ, ε) are unavailable

$$D_{AB} = \frac{10^{-3} T^{1.75} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2}}{P \left[(\Sigma v)_A^{1/3} + (\Sigma v)_B^{1/3} \right]^2}$$

D_{AB} : diffusivity (cm²/s)

T : temperature (K)

P : pressure (atm)

v : diffusion volume

Gas Mass Diffusivity

Fuller Correlation

Atomic and structure diffusion-volume increments, v					
C	16.5	Cl		19.5	
H	1.98	S		17.0	
O	5.48	Aromatic ring		-20.2	
N	5.69	Heterocyclic ring		-20.2	
Diffusion volumes for simple molecules, v					
H ₂	7.07	Ar	16.1	H ₂ O	12.7
D ₂	6.70	Kr	22.8	CClF ₂	114.8
He	2.88	CO	18.9	SF ₆	69.7
N ₂	17.9	CO ₂	26.9	Cl ₂	37.7
O ₂	16.6	N ₂ O	35.9	Br ₂	67.2
Air	20.1	NH ₃	14.9	SO ₂	41.1

If diffusion volume of molecule is not in the table, it can be calculated using the atomic diffusion volume.

Example 2

Determine the diffusion coefficient of benzene in air at 30 °C and 1 atm using Fuller equation.

Gas Mass Diffusivity

Diffusivity for Polar Compound

- Hirschfelder equation is valid but collision integral need to estimate in different way
- Example of polar gas: NH_3 , SO_2 , H_2S
- Collision integral is evaluated by

$$\Omega_D = \Omega_{Do} + \frac{0.196\delta_{AB}^2}{T^*}$$

**Brokaw
Equation**

$$\delta_{AB} = (\delta_A \delta_B)^{\frac{1}{2}}$$

$$\delta = \frac{1.94 \times 10^3 \mu_p^2}{V_b T_b}$$

μ_p : dipole moment (Debye)

V_b : liquid molar volume of compound at its boiling point (cm^3/mol)

T_b : normal boiling point (K)

Gas Mass Diffusivity

Diffusivity for Polar Compound

$$T^* = \kappa T / \varepsilon_{AB}$$



$$\frac{\varepsilon_{AB}}{\kappa} = \left(\frac{\varepsilon_A}{\kappa} \frac{\varepsilon_B}{\kappa} \right)^{1/2}$$

$$\varepsilon / \kappa = 1.18(1 + 1.3 \delta^2) T_b$$

$$\sigma_{AB} = (\sigma_A \sigma_B)^{1/2}$$



$$\sigma = \left(\frac{1.585 V_b}{1 + 1.3 \delta^2} \right)^{1/3}$$

$$\Omega_{D_0} = \frac{A}{(T^*)^B} + \frac{C}{\exp(DT^*)} + \frac{E}{\exp(FT^*)} + \frac{G}{\exp(HT^*)}$$

A = 1.06036	E = 1.03587
B = 0.15610	F = 1.52996
C = 0.19300	G = 1.76474
D = 0.47635	H = 3.89411

Example 3

An absorption tower is proposed to remove selectively ammonia from an exhaust gas stream. Estimate the diffusivity of ammonia in air at 101325 Pa and 373 K using the Brokaw equation. Boiling point for ammonia and air is 239.81 K and 78.8 K. The dipole moment for ammonia is 1.46 debye. Assume the dipole moment of air is zero.

Table 24.4 Molecular volumes at normal boiling point for some commonly encountered compounds

Compound	Molecular volume, in cm ³ /g mol	Compound	Molecular volume, in cm ³ /g mol
Hydrogen, H ₂	14.3	Nitric oxide, NO	23.6
Oxygen, O ₂	25.6	Nitrous oxide, N ₂ O	36.4
Nitrogen, N ₂	31.2	Ammonia, NH ₃	25.8
Air	29.9	Water, H ₂ O	18.9
Carbon monoxide, CO	30.7	Hydrogen sulfide, H ₂ S	32.9
Carbon dioxide, CO ₂	34.0	Bromine, Br ₂	53.2
Carbonyl sulfide, COS	51.5	Chlorine, Cl ₂	48.4
Sulfur dioxide, SO ₂	44.8	Iodine, I ₂	71.5

Gas Mass Diffusivity

Diffusivity in Multicomponent Gas Mixture

$$D_{1-\text{mixture}} = \frac{1}{y'_2/D_{1-2} + y'_3/D_{1-3} + \cdots + y'_n/D_{1-n}}$$

**Wilke
Equation**

$D_{1-\text{mixture}}$: mass diffusivity for component 1 in the gas mixture

D_{1-n} : mass diffusivity of component 1 diffusing through component n

y'_n : mole fraction of component n in gas mixture evaluated on component 1-free basis

$$y'_2 = \frac{y_2}{y_2 + y_3 + \cdots y_n} = \frac{y_2}{1 - y_1}$$

Example 4

In the chemical vapor deposition of silane (SiH_4) on a silicon wafer, a process gas stream rich in an inert nitrogen carrier gas has the following composition:

$$y_{\text{SiH}_4} = 0.0075, \quad y_{\text{H}_2} = 0.015, \quad y_{\text{N}_2} = 0.9775$$

The gas mixture is maintained at 900 K and 100 Pa total system pressure. Determine the diffusivity of silane through the gas mixture. The Lennard-Jones constants for silane are $\epsilon_A/K = 207.6 \text{ K}$ and $\sigma_A = 4.08 \text{ \AA}$

Take Home Message

- $D_{AB} = D_{BA}$ (for system of constant molar concentration)

Self diffusion coefficient

$$D_{AA*} = \frac{2T^{\frac{3}{2}}}{3\pi^{\frac{3}{2}}\sigma_A^2 P} \left(\frac{\kappa^3 N}{M_A} \right)^{\frac{1}{2}}$$

Diffusion coefficient for binary mixture

$$D_{AB} = \frac{2}{3} \left(\frac{\kappa}{\pi} \right)^{\frac{3}{2}} N^{\frac{1}{2}} T^{\frac{3}{2}} \frac{\left(\frac{1}{2M_A} + \frac{1}{2M_B} \right)^{\frac{1}{2}}}{P \left(\frac{\sigma_A + \sigma_B}{2} \right)^2}$$

* No interaction between molecules

Take Home Message

Hirschfelder Equation

$$D_{AB} = \frac{0.001858 T^{\frac{3}{2}} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{\frac{1}{2}}}{P \sigma_{AB}^2 \Omega_D}$$

- Consider interaction between molecules
- For non-polar, non-reacting gas system

Fuller Correlation

$$D_{AB} = \frac{10^{-3} T^{1.75} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2}}{P \left[(\Sigma v)_A^{1/3} + (\Sigma v)_B^{1/3} \right]^2}$$

- When Lennard Jones parameter is not available

Take Home Message

Hirschfelder Equation + Brokaw Equation

$$D_{AB} = \frac{0.001858 T^{\frac{3}{2}} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{\frac{1}{2}}}{P \sigma_{AB}^2 \Omega_D}$$

- For polar system

$$\Omega_D = \Omega_{D0} + \frac{0.196 \delta_{AB}^2}{T^*}$$

Diffusivity in Multicomponent Gas Mixture

$$D_{1-\text{mixture}} = \frac{1}{y'_2/D_{1-2} + y'_3/D_{1-3} + \cdots + y'_n/D_{1-n}}$$